



VISAYAS
STATE UNIVERSITY



Lesson 5.1 - Linear and Nonlinear System of Equations

Department of Mathematics
Visayas State University



At the end of the lesson, the students will be able to:

- Define a linear system of equations.
- Solve a linear system of equations using different methods.
- Define a nonlinear system of equations.
- Solve a nonlinear system of equations.

Linear System of Equations

Up to this point, most problems have involved either an equation in one variable or a single equation in two variables. However, many problems in science, business, and engineering involve two or more equations in two or more variables.

Definition

A **system of equations** is a set of equations that involve the same variables. A **system of linear equations** is a system of equations in which each equation is linear.

To solve such problems, you need to find solutions to the system of equations. Here is an example of a system of two equations in two unknowns.

$$\begin{cases} 2x + y &= 5 \\ 3x - 2y &= 4 \end{cases}$$

Definition

A **solution** of this system is an ordered pair that satisfies each equation in the system. Finding the set of all solutions is called **solving the system of equations**.

Linear System of Equations

For instance, the ordered pair $(2, 1)$ is a solution of the system given in the previous slide. To check this, you can substitute 2 for x and 1 for y in each equation.

In this section, you will study four ways to solve systems of equations namely: graphical method, substitution method, elimination method, and the Cramer's rule.

Example (Graphical Method)

Solve the system by graphic method,

$$\begin{cases} x + y = 6 \\ x - y = 2 \end{cases}.$$

Linear System of Equations

The Method of Substitution

- 1 **Solve** one of the equations for one variable in terms of the other.
- 2 **Substitute** the expression found in Step 1 into the other equation to obtain an equation in one variable.
- 3 **Solve** the equation obtained in Step 2.
- 4 **Back - substitute** the value obtained in Step 3 into the expression obtained in Step 1 to find the value of the other variable.
- 5 Check that the solution satisfies **each** of the original equations.

Example

$$\textcircled{1} \begin{cases} x + y = 4 \\ x - y = 2 \end{cases}$$

$$\textcircled{2} \begin{cases} 2x - 3y = -7 \\ 3x + y = -5 \end{cases}$$

$$\textcircled{3} \begin{cases} 5x + 3y = 9 \\ 2x - 4y = 14 \end{cases}$$

Linear System of Equations

The Method of Elimination

- 1 Obtain **coefficients** for x (or y) that differ only in sign by multiplying all terms of one or both equations by suitably chosen constants.
- 2 **Add** the equations to eliminate one variable, and solve the resulting equation.
- 3 **Back - substitute** the value obtained in Step 2 into either of the original equations and solve for the other variable.
- 4 Check your solution in **both** of the original equations.

Example

$$\textcircled{1} \begin{cases} 4x + 5y = 6 \\ 2x + 3y = 1 \end{cases}$$

$$\textcircled{2} \begin{cases} 5x + 3y = 0 \\ 8x - y = 29 \end{cases}$$

$$\textcircled{3} \begin{cases} 0.02x - 0.05y = -0.38 \\ 0.03x + 0.04y = 1.04 \end{cases}$$

Linear System of Equations

Example (Other cases of the solutions)

$$\textcircled{1} \begin{cases} x - 2y = 3 \\ -2x + 4y = 1 \end{cases}$$

$$\textcircled{2} \begin{cases} 2x - y = 1 \\ 4x - 2y = 2 \end{cases}$$

Graphical Interpretation of Solutions

A general system of equations can have exactly one solution, two or more solutions, or no solution. If a system of linear equations has two different solutions, it must have an infinite number of solutions.

Graphical Interpretations of Solutions

For a system of two linear equations in two variables, the number of solutions is one of the following.

<i>Number of Solutions</i>	<i>Graphical Interpretation</i>	<i>Slopes of Lines</i>
1. Exactly one solution	The two lines intersect at one point.	The slopes of the two lines are not equal.
2. Infinitely many solutions	The two lines coincide (are identical).	The slopes of the two lines are equal.
3. No solution	The two lines are parallel.	The slopes of the two lines are equal.

Linear System of Equations

Example (Linear System of Equations in three variables)

$$\textcircled{1} \begin{cases} x + y + z &= 2 \\ 2x - 2y - z &= 2 \\ x + 2y - z &= -3 \end{cases}$$

$$\textcircled{2} \begin{cases} x - 2y + 3z &= 9 \\ -x + 3y &= -4 \\ 2x - 5y + 5z &= 17 \end{cases}$$

Example (Nonsquare System)

Solve the system of linear equations. $\begin{cases} x - 2y + z = 2 \\ 2x - y - z = 1 \end{cases}$

Linear System of Equations

Definition (The Cramer's Rule)

Cramer's rule is one of the important methods applied to solve a system of equations. In this method, the values of the variables in the system are to be calculated using the determinants of matrices. Thus, Cramer's rule is also known as the determinant method.

Consider a system of two equations in two variables.

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Then, $x = \frac{D_x}{D}$ and $y = \frac{D_y}{D}$ where D , D_x and D_y are determinants given by

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1,$$

$$D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} = c_1b_2 - c_2b_1,$$

$$D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} = a_1c_2 - a_2c_1$$

Linear System of Equations

Example

$$\textcircled{1} \begin{cases} 4x + 5y = 6 \\ 2x + 3y = 1 \end{cases}$$

$$\textcircled{2} \begin{cases} 5x + 3y = 0 \\ 8x - y = 29 \end{cases}$$

$$\textcircled{3} \begin{cases} x + y + z = 2 \\ 2x - 2y - z = 2 \\ x + 2y - z = -3 \end{cases}$$

$$\textcircled{4} \begin{cases} x - 2y + 3z = 9 \\ -x + 3y = -4 \\ 2x - 5y + 5z = 17 \end{cases}$$

Linear System of Equations

Example (Applications)

- ① A total of Php600,000 is invested in two funds paying 5% and 3% simple interest. (Recall that the formula for simple interest is $I = Prt$ where P is the principal, r is the annual interest rate, and t is the time.) The yearly interest is Php25,000. How much is invested at each rate?
- ② The weekly ticket sales for a new comedy movie decreased each week. At the same time, the weekly ticket sales for a new drama movie increased each week. Models that approximate the weekly ticket sales S (in millions of dollars) for each movie are
$$\begin{cases} S = 60 - 8x & \text{Comedy} \\ S = 10 + 4.5x & \text{Drama} \end{cases}$$
where x represents the number of weeks each movie was in theaters, with $x = 0$ corresponding to the ticket sales during the opening weekend. After how many weeks will the ticket sales for the two movies be equal?

Linear System of Equations

Example (Applications)

An airplane flying into a headwind travels the 2000-mile flying distance between Chicopee, Massachusetts, and Salt Lake City, Utah in 4 hours and 24 minutes. On the return flight, the same distance is traveled in 4 hours. Find the airspeed of the plane and the speed of the wind, assuming that both remain constant.

The total cost C of producing x units of a product typically has two components - the initial cost and the cost per unit. When enough units have been sold so that the total revenue R equals the total cost C , the sales are said to have reached the **break-even point**. You will find that the break-even point corresponds to the point of intersection of the cost and revenue curves.

Example (Applications)

A shoe company invests Php15,000,000 in equipment to produce a new line of athletic footwear. Each pair of shoes costs Php250 to produce and is sold for Php3,000. How many pairs of shoes must be sold before the business breaks even?

Nonlinear System of Equations

The method of substitution can also be used to solve systems in which one or both of the equations are nonlinear.

Example

$$\textcircled{1} \begin{cases} x^2 + 4x - y = 7 \\ 2x - y = -1 \end{cases}$$

$$\textcircled{2} \begin{cases} -x + y = 4 \\ x^2 + y = 3 \end{cases}$$

$$\textcircled{3} \begin{cases} y = -x \\ x^2 + y^2 = 8 \end{cases}$$